

The Poisson Distribution

ANSWER KEY



Conditions for a Poisson distribution

- 1 State the key conditions required for a random variable to be modeled by a Poisson distribution.
Events occur independently of one another; events occur at a constant average rate over the given interval (of time or space); two events cannot occur at exactly the same instant; the number of events in non-overlapping intervals is independent.

Computing Poisson probabilities

- 2 A call center receives an average of 4 calls per minute, modeled by a Poisson distribution with $\lambda = 4$. Find $P(X = 6)$, using $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$, to three decimal places.
 $P(X = 6) = e^{-4} 4^6 / 6! = (0.0183)(4096) / 720 \approx 74.98 / 720 \approx 0.104$.
- 3 Using the same call center ($\lambda = 4$ calls per minute), find the probability of receiving at most 2 calls in a given minute.
 $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$. $P(X = 0) = e^{-4} \approx 0.0183$. $P(X = 1) = 4e^{-4} \approx 0.0733$.
 $P(X = 2) = 16e^{-4} / 2 = 8e^{-4} \approx 0.1465$. Sum $\approx 0.0183 + 0.0733 + 0.1465 \approx 0.238$.

Mean and variance of a Poisson distribution

- 4 State the mean and variance of a Poisson distribution with parameter λ , and explain what is distinctive about this relationship compared to other common distributions.
For a Poisson distribution, both the mean and the variance equal λ : $E(X) = \lambda$ and $\text{Var}(X) = \lambda$. This is distinctive because for most other distributions, the mean and variance are independent parameters (e.g. the normal distribution has separate μ and σ^2) — for Poisson, a single parameter determines both.

Sums of independent Poisson variables

- 5 Two independent Poisson processes have rates $\lambda_1 = 3$ and $\lambda_2 = 5$ events per hour. If $X_1 \sim \text{Po}(3)$ and $X_2 \sim \text{Po}(5)$, state the distribution of $X_1 + X_2$, and find its mean and variance.
The sum of independent Poisson random variables is itself Poisson: $X_1 + X_2 \sim \text{Po}(3 + 5) = \text{Po}(8)$. Mean = 8; variance = 8 (since for Poisson, mean and variance are equal).

Poisson as an approximation to the binomial

- 6 Explain the conditions under which a binomial distribution $X \sim B(n, p)$ can be well-approximated by a Poisson distribution, and state the appropriate value of λ to use.
- The Poisson approximation to the binomial is appropriate when n is large and p is small (a common guideline: $n > 50$ and $np < 5$), so that events are rare relative to the number of trials. In that case, X is approximately $Po(\lambda)$ with $\lambda = np$.
- 7 A factory produces 2000 items, each independently defective with probability 0.002. Using the Poisson approximation, find the probability that exactly 3 items are defective, to three decimal places.
- $\lambda = np = 2000(0.002) = 4$. $P(X = 3) = e^{-4}4^3/3! = (0.0183)(64)/6 \approx 1.171/6 \approx 0.195$.
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Using cumulative Poisson probabilities

- 8 A shop receives customers at an average rate of $\lambda = 5$ per 10-minute interval. Find the probability of receiving more than 2 customers in a given 10-minute interval, given $P(X = 0) \approx 0.0067$, $P(X = 1) \approx 0.0337$, and $P(X = 2) \approx 0.0842$.
- $P(X > 2) = 1 - P(X \leq 2)$. Summing the three given probabilities:
 $P(X \leq 2) = 0.0067 + 0.0337 + 0.0842 = 0.1246$. So $P(X > 2) = 1 - 0.1246 = 0.8754$.
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Finding an unknown parameter from a given probability

- 9 A random variable $X \sim Po(\lambda)$ satisfies $P(X = 0) = 0.1353$. Find λ , to the nearest whole number, using $P(X = 0) = e^{-\lambda}$.
- $e^{-\lambda} = 0.1353$. Taking natural logs: $-\lambda = \ln(0.1353) \approx -2.0$, so $\lambda \approx 2$.
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Variance-based reasoning with the Poisson distribution

- 10 A random variable $X \sim Po(\lambda)$ has standard deviation 3. Find λ , and then find $P(X = \lambda)$, to three decimal places.
- Since variance $= \lambda$ and standard deviation $= \sqrt{\lambda} = 3$, we get $\lambda = 9$.
- $P(X = 9) = e^{-9}9^9/9! = (0.0001234)(387420489)/362880 \approx 47790/362880 \approx 0.132$.
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Combining Poisson reasoning with real-world decisions

- 11 A website server crashes on average $\lambda = 0.5$ times per week. The IT team wants to know the probability of at least one crash in a given week, given $P(X = 0) \approx 0.6065$. Should the team be concerned if their target is to keep this probability below 30%?
- $P(X \geq 1) = 1 - P(X = 0) = 1 - 0.6065 = 0.3935$, or about 39.35%. Since this exceeds their target threshold of 30%, the team should be concerned — there is roughly a 39% chance of at least one crash in a given week, above their comfort threshold, suggesting they should investigate ways to reduce the average crash rate λ .

Combining a Poisson approximation with conditional probability

- 12** Machine faults occur at an average rate of $\lambda = 1.5$ per day. Find the probability that exactly one fault occurs today, given that at least one fault occurs today, to three decimal places, using $P(X = 0) \approx 0.2231$ and $P(X = 1) \approx 0.3347$.

$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.2231 = 0.7769$. Conditional probability:

$$P(X = 1 \mid X \geq 1) = P(X = 1)/P(X \geq 1) = 0.3347/0.7769 \approx 0.431.$$

Combining Poisson processes over different time intervals

- 13** A radioactive source emits particles at a rate of $\lambda = 2$ per second, modeled by a Poisson distribution. Find $P(X = 3)$ for a 3-second interval, to three decimal places, using the fact that the parameter scales with the length of the interval.

For a 3-second interval, the parameter becomes $\lambda = 2 \times 3 = 6$.

$$P(X = 3) = e^{-6} 6^3 / 3! = (0.002479)(216)/6 \approx 0.5354/6 \approx 0.089.$$

- 14** Explain why the Poisson distribution is not an appropriate model for the number of heads obtained when flipping a fair coin 10 times, referencing the conditions required for a Poisson model.

The number of heads in 10 coin flips is naturally modeled by a binomial distribution $B(10, 0.5)$, not a Poisson distribution, because a Poisson model requires events to be rare (a small probability of occurrence per trial, with np moderate while n is large) — here $p = 0.5$ is far too large for the Poisson approximation to be valid, and there is a fixed, small number of trials ($n = 10$) rather than events occurring continuously over time or space.

Comparing two independent Poisson processes

- 15** Two independent Poisson processes have rates $\lambda_A = 3$ and $\lambda_B = 2$ per hour. Find the probability that, in a given hour, process A produces exactly 2 events and process B produces exactly 1 event, to three decimal places, given $P(X_A = 2) \approx 0.2240$ and $P(X_B = 1) \approx 0.2707$.

Since the two processes are independent, the joint probability is the product of the individual probabilities:

$$P(X_A = 2) \times P(X_B = 1) = 0.2240 \times 0.2707 \approx 0.0606.$$

Rescaling the Poisson parameter for a different interval

- 16** A radio station receives an average of 12 requests per hour. Find the appropriate Poisson parameter λ to use for a 15-minute segment.

Since the rate is 12 per hour and 15 minutes is a quarter of an hour, the parameter for that segment is $\lambda = 12/4 = 3$.

The mode of a Poisson distribution

- 17** A random variable $X \sim \text{Po}(4)$ has an integer parameter $\lambda = 4$. State the mode(s) of X (the most likely value or values), and explain why a Poisson distribution with an integer λ has two modes, using the ratio $P(X = k)/P(X = k - 1) = \lambda/k$.

At $k = \lambda$, the ratio $P(X = k)/P(X = k - 1) = \lambda/k = \lambda/\lambda = 1$, so $P(X = \lambda) = P(X = \lambda - 1)$ exactly. This means both $X = 3$ and $X = 4$ share the same (maximum) probability, so the distribution has two modes: 3 and 4. This only happens when λ is a positive integer; for a non-integer λ , there is a single mode at $\lfloor \lambda \rfloor$.

Identifying an appropriate Poisson scenario

- 18** Which of the following scenarios is best modeled by a Poisson distribution?

- a.** The number of typing errors on a randomly selected page of a manuscript. ✓
- b.** The number of heads in 20 coin tosses.
- c.** The height of a randomly selected adult.
- d.** The number of correct answers on a 10-question multiple-choice test, each with 4 options.

Typing errors occur rarely and independently across a fixed amount of text (a page), matching the Poisson conditions of a constant average rate with no fixed number of trials. The coin-toss and multiple-choice scenarios are binomial (a fixed number of trials, each with the same success probability), and height is a continuous variable following approximately a normal distribution — neither fits the Poisson framework.

Assessing Poisson plausibility from sample data

- 19** A biologist counts the number of bacteria colonies in 10 equal-sized petri dishes: 3, 5, 4, 6, 5, 4, 3, 5, 6, 4. Find the sample mean and the unbiased sample variance, and comment on whether a Poisson model seems plausible for this data.

Mean = $(3 + 5 + 4 + 6 + 5 + 4 + 3 + 5 + 6 + 4)/10 = 45/10 = 4.5$. Squared deviations from the mean sum to $4(2.25) + 6(0.25) = 9 + 1.5 = 10.5$ (the values 3 and 6 each deviate by 1.5, contributing 2.25 each, four times in total; the values 4 and 5 each deviate by 0.5, contributing 0.25 each, six times in total). Unbiased sample variance: $10.5/(10 - 1) = 10.5/9 \approx 1.167$. Since the sample variance (≈ 1.17) is much smaller than the sample mean (4.5), a Poisson model seems implausible for this data — a genuine Poisson distribution requires the mean and variance to be approximately equal, but this data shows far less spread than a Poisson model would predict (a pattern called underdispersion).